COURSE 5 · ASTRONOMY

Astronomy

Printable flashcards

154 cards: equations, key ideas, quick-check questions and problems.
Cut, fold and review; or use the scheduled version in the app.

How to use

Printable flashcards

Each row is one card: the prompt on the left, the answer on the right. Print single-sided, cut along the rows, and fold each strip down the middle so the answer sits on the back. The same cards are in the app, where they are scheduled for review automatically.

Module 5.1 · The sky and the history of astronomy

EQUATION Write the equation: Earth's axial tilt	$\varepsilon \approx 23.4^\circ$
EQUATION Write the equation: Noon Sun altitude at the solstices	$h = 90^\circ - \lvert\text{latitude}\rvert \pm 23.4^\circ$
EQUATION Write the equation: Synodic month	$P_{\text{phases}} \approx 29.5 \text{ days}$
KEY IDEA Complete the key idea: Seasons come from …	Seasons come from axial tilt, not distance
KEY IDEA Complete the key idea: Phases: how much of …	Phases: how much of the Moon's lit half we see
KEY IDEA Complete the key idea: Southern sky: Southern …	Southern sky: Southern Cross, Magellanic Clouds, galactic centre
KEY IDEA Complete the key idea: Copernicus, Tycho, Kepler and …	Copernicus, Tycho, Kepler and Galileo moved the Sun to the centre
QUICK CHECK Earth's axial tilt is about	23.4°
QUICK CHECK The Magellanic Clouds are	nearby galaxies
QUICK CHECK Copernicus put at the centre	the Sun

QUICK CHECK The Moon's phase cycle is about	29.5 days
PROBLEM What causes Earth's seasons?	the tilt of Earth's axis. Tilt changes how directly sunlight strikes each hemisphere.
PROBLEM Why does the Moon show phases?	we see different amounts of its sunlit half. Half is always lit; our viewing angle changes.
PROBLEM Brisbane is at latitude 27.5° S. What is the Sun's noon altitude at the December (summer) solstice, in degrees?	85.9 °. ≈ 85.9°, almost overhead.
PROBLEM And at the June (winter) solstice?	39.1 °. h = 90 − 27.5 − 23.4 ≈ 39.1°.
PROBLEM In which month is Earth closest to the Sun?	January. Perihelion is around 3–4 January.
PROBLEM Who first showed planetary orbits are ellipses?	Kepler. Kepler, using Tycho's data, 1609.
PROBLEM Galileo's observation that Venus shows a full set of phases supported	the Sun-centred model. Full phases require Venus to orbit the Sun.
PROBLEM Why can't people in Europe see the Southern Cross?	The Southern Cross lies close to the south celestial pole. From northern latitudes, that part of the celestial sphere stays permanently below the horizon as Earth rotates, so it never rises. You need to be roughly south of about 25° N to see it.

Module 5.2 · Gravity and orbits

EQUATION Write the equation: Kepler's third law (Solar System units)	$T^2 = a^3 \quad (T \text{ in years},\ a \text{ in AU})$
EQUATION Write the equation: Universal gravitation	$F = \frac{GMm}{r^2}, \quad G = 6.67 \times 10^{-11}\ \text{N m}^2\,\text{kg}^{-2}$

EQUATION Write the equation: Surface gravity	$g = \frac{GM}{R^2}$
EQUATION Write the equation: Circular orbital speed	$v = \sqrt{\frac{GM}{r}}$
EQUATION Write the equation: Kepler's third law (Newton's form)	$T^2 = \frac{4\pi^2}{GM} r^3$
EQUATION Write the equation: Escape velocity	$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$
KEY IDEA Complete the key idea: Kepler: ellipses, equal …	Kepler: ellipses, equal areas, $T^2 \propto a^3$
KEY IDEA Complete the key idea: Newton: F = …	Newton: $F = GMm/r^2$, an inverse-square law
KEY IDEA Complete the key idea: Orbital speed v = …	Orbital speed $v = \sqrt{(GM/r)}$, independent of the orbiting mass
KEY IDEA Complete the key idea: Measuring an orbit …	Measuring an orbit weighs the central body
QUICK CHECK Gravitational force between two masses varies as	$1/r^2$
QUICK CHECK Kepler's first law says orbits are	ellipses
QUICK CHECK Orbital speed depends on	the central mass and distance
QUICK CHECK Cavendish's experiment measured	G
PROBLEM A planet orbits at 4 AU from the Sun. What is its period in years?	8 years. $T^2 = 4^3 = 64$.

PROBLEM If the distance between two masses triples, by what factor does the gravitational force change? Give a decimal.	0.1111. Inverse square: $1/3^2 = 1/9 \approx 0.111$.
PROBLEM Mars has mass 6.42×10^{23} kg and radius 3.39×10^6 m. What is its surface gravity?	3.726 m/s². ≈ 3.7 m/s².
PROBLEM The ISS orbits at r = 6.78×10^6 m from Earth's centre. Earth's GM = 3.99×10^{14} m³/s². What is its orbital speed?	7671 m/s. ≈ 7670 m/s.
PROBLEM Astronauts in the ISS float because	they are in free fall with the station. Gravity there is about 90% of surface strength.
PROBLEM What is the escape velocity from the Moon? (GM = 4.9×10^{12} m³/s², R = 1.74×10^6 m)	2373 m/s. ≈ 2370 m/s.
PROBLEM By Kepler's second law, a comet moves fastest	closest to the Sun. Equal areas in equal times.

Module 5.3 · Light as messenger

EQUATION Write the equation: Inverse-square law	$b = \frac{L}{4\pi d^2}$
EQUATION Write the equation: Parallax distance	$d\,(\text{pc}) = \frac{1}{p\,(\text{arcsec})}, \quad 1\ \text{pc} \approx 3.26\ \text{ly}$
EQUATION Write the equation: Wien's law	$\lambda_{\max} = \frac{2.90 \times 10^{-3}\ \text{m K}}{T}$
EQUATION Write the equation: Doppler shift (v much less than c)	$\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$
KEY IDEA Complete the key idea: Telescopes go to space …	Telescopes go to space for wavelengths the atmosphere blocks

KEY IDEA Complete the key idea: Brightness ∝ 1/distance^2; parallax …	Brightness ∝ 1/distance^2; parallax gives distances to nearby stars
KEY IDEA Complete the key idea: Hotter means …	Hotter means bluer: λ_max ∝ 1/T
KEY IDEA Complete the key idea: Spectral lines reveal …	Spectral lines reveal composition; their shift reveals motion
QUICK CHECK A redshift means the source is	receding
QUICK CHECK Hotter stars appear	bluer
QUICK CHECK 1 parsec is about	3.26 light-years
QUICK CHECK Absorption lines tell us a star's	composition
PROBLEM A star is moved to three times its distance. By what factor does its apparent brightness change? Give a decimal.	0.1111. Inverse square: 1/9.
PROBLEM A star has a parallax of 0.25 arcseconds. How far away is it, in parsecs?	4 pc. d = 1/p = 1/0.25.
PROBLEM Convert 4 parsecs to light-years.	13.04 ly. 4 × 3.26 ≈ 13 ly.
PROBLEM The Sun's surface is about 5800 K. What is its peak wavelength in nanometres?	500 nm. = 500 nm, green-blue.
PROBLEM A star's peak wavelength is 290 nm. What is its surface temperature?	1e+04 K. T = 2.90×10^{-3} / 2.90×10^{-7} = 10 000 K.

PROBLEM A hydrogen line normally at 656.3 nm is observed at 656.9 nm. How fast is the star receding, in km/s? ($c = 3.0 \times 10^5$ km/s)	274.3 km/s. $v = 9.14 \times 10^{-4} \times 3 \times 10^5 \approx 274$ km/s.
PROBLEM Why are X-ray telescopes placed in space?	Earth's atmosphere absorbs X-rays. The atmosphere is opaque to X-rays.
PROBLEM How did astronomers know helium existed before it was found on Earth?	In 1868, during a solar eclipse, a bright spectral line was seen in the Sun's light at a wavelength that matched no known element. Since every element has a unique set of lines, it had to be a new element, named after the Greek Helios. It was isolated on Earth in 1895.

Module 5.4 · The Solar System

EQUATION Write the equation: Solar power from mass loss	$P = \frac{\Delta m}{\Delta t} c^2 \approx 3.8 \times 10^{26}$ W
EQUATION Write the equation: Astronomical unit	$1 \text{ AU} \approx 1.50 \times 10^{11}$ m
EQUATION Write the equation: Transit depth	$\frac{\Delta F}{F} \approx \left(\frac{R_{\text{planet}}}{R_{\text{star}}}\right)^2$
KEY IDEA Complete the key idea: The Sun is a fusion-powered …	The Sun is a fusion-powered ball of plasma with 99% of the mass
KEY IDEA Complete the key idea: Inner planets rocky, outer …	Inner planets rocky, outer planets giants: the frost line explains it
KEY IDEA Complete the key idea: The Solar System formed from …	The Solar System formed from a spinning disc 4.6 billion years ago
KEY IDEA Complete the key idea: Exoplanets: transits …	Exoplanets: transits and Doppler wobbles

QUICK CHECK The Sun contains about what share of the Solar System's mass?	over 99%
QUICK CHECK The Solar System is about how old?	4.6 billion years
QUICK CHECK The transit method detects planets by	starlight dimming
QUICK CHECK Uranus and Neptune are	ice giants
PROBLEM What powers the Sun?	nuclear fusion. Hydrogen fuses into helium in the core.
PROBLEM Jupiter is 5.2 AU from the Sun. How many kilometres is that? (1 AU = 1.5×10^8 km)	7.800e+08 km. $5.2 \times 1.5 \times 10^8 = 7.8 \times 10^8$ km.
PROBLEM How long does sunlight take to reach Neptune, 30 AU away, in hours? (Earth: 500 s)	4.167 h. ÷ 3600 ≈ 4.2 h.
PROBLEM Why are the inner planets rocky and the outer ones gas giants?	near the Sun it was too hot for ices to condense. The frost line.
PROBLEM Why do the planets orbit in nearly the same plane?	they formed from a flattened spinning disc. Conservation of angular momentum flattened the cloud.
PROBLEM A Jupiter-sized planet (radius 0.1 × the star's) transits its star. By what fraction does the starlight dim? Give a decimal.	0.01. $(0.1)^2 = 0.01$, a 1% dip.
PROBLEM Why is Venus hotter than Mercury even though it is further from the Sun?	Venus has a very thick carbon dioxide atmosphere. Sunlight warms the surface, which radiates infrared; the CO_2 traps that infrared (a runaway greenhouse effect), raising the surface to about 465 °C. Mercury has almost no atmosphere, so its heat escapes and its night side gets very cold.

Module 5.5 · Stars

EQUATION Write the equation: Stefan–Boltzmann law	$L = 4\pi R^2 \sigma T^4, \quad \sigma = 5.67 \times 10^{-8}\ \mathrm{W\,m^{-2}K^{-4}}$
EQUATION Write the equation: Main-sequence lifetime	$t \approx 10^{10} \left(\frac{M}{M_\odot}\right)^{-2.5}\ \text{years}$
EQUATION Write the equation: Hydrogen fusion (net)	$4\,{}^1\mathrm{H} \rightarrow {}^4\mathrm{He} + \text{energy}$
KEY IDEA Complete the key idea: Stars balance gravity …	Stars balance gravity against pressure from fusion energy
KEY IDEA Complete the key idea: L = 4πR²σT⁴: luminosity …	L = 4πR²σT⁴: luminosity depends on size and temperature⁴
KEY IDEA Complete the key idea: HR diagram: main …	HR diagram: main sequence, giants, white dwarfs
KEY IDEA Complete the key idea: More massive stars are …	More massive stars are brighter and live much shorter lives
QUICK CHECK Stars are powered by	fusion
QUICK CHECK On an HR diagram, hot stars are traditionally on the	left
QUICK CHECK Luminosity depends on temperature to the power	4
QUICK CHECK The most massive stars have	the shortest lives
PROBLEM Two stars are the same size, but one is twice as hot. How many times more luminous is it?	16 ×. 2^4 = 16.

PROBLEM A star has the same temperature as the Sun but 10 times its radius. How many times more luminous is it?	100 ×. $10^2 = 100$.
PROBLEM On the HR diagram, a hot but very faint star is most likely a	white dwarf. Hot and faint means small.
PROBLEM Using t ≈ 10^{10} × M^(−2.5) years, estimate the lifetime of a star of 4 solar masses.	3.125e+08 years. $10^{10} \div 32 \approx 3.1 \times 10^8$ years.
PROBLEM What keeps a stable star from collapsing?	pressure from hot gas and radiation. Pressure balances gravity.
PROBLEM In hydrogen fusion about 0.7% of mass becomes energy. How much energy comes from fusing 1 kg of hydrogen? ($c = 3 \times 10^8$ m/s)	6.300e+14 J. $E = 0.007 \times 9 \times 10^{16} = 6.3 \times 10^{14}$ J.
PROBLEM Where do stars spend most of their lives?	on the main sequence. Fusing hydrogen in the core.
PROBLEM Massive stars have much more fuel than the Sun. Why do they live much shorter lives?	Luminosity rises far faster than mass, roughly as M^3.5. A star ten times the Sun's mass has ten times the fuel but burns it thousands of times faster, so it runs out in a fraction of the time: lifetime goes roughly as $M/L \propto$ M^−2.5.

Module 5.6 · Stellar death

EQUATION Write the equation: Schwarzschild radius	$R_s = \frac{2GM}{c^2}$
EQUATION Write the equation: Density	$\rho = \frac{M}{\frac{4}{3}\pi R^3}$
EQUATION Write the equation: Chandrasekhar limit (white dwarf maximum)	$M_{\max} \approx 1.4\, M_\odot$

KEY IDEA Complete the key idea: Sun-like stars: red …	Sun-like stars: red giant, planetary nebula, white dwarf
KEY IDEA Complete the key idea: Massive stars: …	Massive stars: iron core collapse, supernova
KEY IDEA Complete the key idea: Heavy elements are …	Heavy elements are made in stars and supernovae
KEY IDEA Complete the key idea: Remnants: white dwarf, neutron …	Remnants: white dwarf, neutron star or black hole, by mass
QUICK CHECK A planetary nebula is	the ejected outer layers of a dying star
QUICK CHECK Neutron stars are about the size of	a city
QUICK CHECK A black hole's boundary is the	event horizon
QUICK CHECK Supernovae occur for stars above about	8. About eight solar masses.
PROBLEM What will the Sun become at the end of its life?	a white dwarf. It is too low in mass for a supernova.
PROBLEM Where were most of the heavy elements in your body made?	inside stars and supernovae. Stellar nucleosynthesis.
PROBLEM What is the Schwarzschild radius of a 10 solar-mass black hole, in km? ($M_\odot = 2.0 \times 10^{30}$ kg, $G = 6.67 \times 10^{-11}$, $c = 3.0 \times 10^{8}$)	29.64 km. ≈ 29 600 m ≈ 30 km.
PROBLEM A white dwarf has mass 1.2×10^{30} kg and radius 6.0×10^{6} m. What is its density, in kg/m^3?	1.326e+09 kg/m^3. $\rho \approx 1.3 \times 10^{9}$ kg/m^3.

PROBLEM Why does fusion stop at iron in massive stars?	fusing iron absorbs rather than releases energy. Iron has the most tightly bound nucleus.
PROBLEM Pulsars are	rotating neutron stars. Discovered by Jocelyn Bell Burnell, 1967.
PROBLEM What is the maximum mass of a white dwarf?	about 1.4 solar masses. The Chandrasekhar limit.
PROBLEM If the Sun were replaced by a black hole of the same mass, what would happen to Earth's orbit, and why?	Nothing would change in the orbit. Gravity outside a spherical mass depends only on the mass and distance ($F = GMm/r^2$), not on how compact the object is. Earth would keep orbiting at the same distance and speed, though it would be dark and cold.

Module 5.7 · Galaxies

EQUATION Write the equation: Orbital period around the galaxy	$T = \frac{2\pi r}{v}$
EQUATION Write the equation: Mass inside an orbit	$M = \frac{v^2 r}{G}$
EQUATION Write the equation: Light-year	$1\ \text{ly} \approx 9.46 \times 10^{15}\ \text{m}$
KEY IDEA Complete the key idea: The Milky Way is a barred …	The Milky Way is a barred spiral; the Sun is 26 000 ly from the centre
KEY IDEA Complete the key idea: Galaxy types: …	Galaxy types: spiral, elliptical, irregular
KEY IDEA Complete the key idea: Supermassive black holes …	Supermassive black holes sit at galactic centres
KEY IDEA Complete the key idea: Flat rotation …	Flat rotation curves reveal dark matter

QUICK CHECK The Milky Way contains roughly how many stars?	hundreds of billions
QUICK CHECK Vera Rubin's work provided evidence for	dark matter
QUICK CHECK Andromeda is	a separate galaxy
QUICK CHECK Our galaxy's central black hole has a mass of about	4 million solar masses
PROBLEM What type of galaxy is the Milky Way?	barred spiral. A barred spiral.
PROBLEM Using $M = v^2 r / G$ with the numbers above, estimate the mass inside the Sun's orbit, in kg.	1.983e+41 kg. $\approx 2.0 \times 10^{41}$ kg, about 10^{11} solar masses.
PROBLEM Flat galaxy rotation curves are evidence for	dark matter. Extra unseen mass in the halo.
PROBLEM Andromeda is 2.5 million light-years away. When we look at it, how long ago did that light leave, in years?	2.500e+06 years. One light-year per year of travel.
PROBLEM Elliptical galaxies typically contain	mostly old stars and little gas. They are red and dead, mostly.
PROBLEM How was the mass of the black hole at the Milky Way's centre measured?	from the orbits of nearby stars. Kepler's third law applied to stars orbiting it.
PROBLEM Why did astronomers expect stars far from a galaxy's centre to orbit more slowly, and what did they find?	If most of the mass were concentrated in the bright centre, orbital speed far out would fall as $v = \sqrt{(GM/r)}$, like the outer planets. Instead, speeds stay roughly constant at large radius, which means the enclosed mass keeps growing with distance even where little light is seen. This implies a large halo of dark matter.

Module 5.8 · Cosmology

EQUATION Write the equation: Hubble's law	$v = H_0 d, \quad H_0 \approx 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$
EQUATION Write the equation: Hubble time	$t \approx \frac{1}{H_0}$
EQUATION Write the equation: Composition today	$\Omega_b \approx 0.05, \quad \Omega_{dm} \approx 0.27, \quad \Omega_\Lambda \approx 0.68$
EQUATION Write the equation: Cosmological redshift	$z = \frac{\lambda_{\mathrm{obs}} - \lambda_{\mathrm{emit}}}{\lambda_{\mathrm{emit}}}$
KEY IDEA Complete the key idea: Hubble's law: v = …	Hubble's law: v = H₀d; space itself is expanding
KEY IDEA Complete the key idea: Age ≈ 1/H₀ …	Age ≈ 1/H₀ ≈ 13.8 billion years
KEY IDEA Complete the key idea: Big Bang evidence: …	Big Bang evidence: expansion, microwave background, light-element abundances
KEY IDEA Complete the key idea: Universe: 5% ordinary matter, …	Universe: 5% ordinary matter, 27% dark matter, 68% dark energy
QUICK CHECK Hubble's law says recession speed is proportional to	distance
QUICK CHECK The universe is about how old?	13.8 billion years
QUICK CHECK The expansion of the universe is	accelerating
QUICK CHECK The CMB has a temperature of about	2.7 K
PROBLEM A galaxy is 100 Mpc away. Using H₀ = 70 km/s/Mpc, how fast is it receding?	7000 km/s. v = H₀d = 70 × 100.

PROBLEM A galaxy recedes at 14 000 km/s. How far away is it?	200 Mpc. $d = v/H_0 = 14\,000/70$.
PROBLEM Estimate the age of the universe as $1/H_0$. $H_0 = 70$ km/s/Mpc, 1 Mpc = 3.09×10^{19} km, 1 year = 3.15×10^7 s. Give billions of years.	14.01 billion years. $\div 3.15 \times 10^7 \approx 1.4 \times 10^{10}$ years.
PROBLEM A line emitted at 500 nm is observed at 600 nm. What is the redshift z?	0.2. $z = (600 - 500)/500$.
PROBLEM The cosmic microwave background is	light released when atoms first formed, now stretched to microwaves. About 380 000 years after the Big Bang.
PROBLEM Which is NOT one of the main pillars of evidence for the Big Bang?	the existence of black holes. Black holes are not evidence for the Big Bang.
PROBLEM Roughly what fraction of the universe is ordinary matter?	5%. The rest is dark matter and dark energy.
PROBLEM If all galaxies are moving away from us, does that mean we are at the centre of the universe? Explain.	No. In a uniformly expanding universe, every observer sees every other galaxy receding, with speed proportional to distance. Like dots on an inflating balloon, no dot is special. The expansion is of space itself, with no centre and no edge.